Predicting Length Of Stay & Readmitted Patients In Hospitals

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***Abstract*—The length of stay (LOS) has a direct bearing on patient outcomes, resource use, and hospital revenue, making it a crucial element in healthcare administration. Accurate LOS prediction can aid hospitals in discharge planning and resource allocation. In recent years, LOS prediction models have been created using machine learning approaches. Given that hospital readmission is linked to higher healthcare costs and poorer patient outcomes, it is a major concern over healthcare providers. Hospitals can target interventions and reduce readmissions by predicting how many patients are in danger of doing so. Considering the data resources used, the machine learning strategies used, and the performance measures employed to assess the models, this work seeks to provide an overview of the most recent state of the art in LOS prediction. We examined the LOS prediction literature from several datasets and determined the most popular methods. A medical history, physical examination, and diagnostic tests like electrocardiograms, echocardiograms, and cardiac catheterization are frequently used to diagnose heart disease. Depending on the kind and severity of the condition, there are different therapies for heart disease, including lifestyle changes, medication, surgery, or a combination of these. LR, KNN , Decision Tree, RF, Naive Bayes.**

***Index Terms*—Logistic regression, Decision trees, Random forests, Naive Bayes,K-Nearest Neighbors Algorithm.**

I. Introduction

An important task in healthcare, particularly in hospital settings, is predicting the length of stay (LOS). A patient's length of stay is the total number of days they are admitted and remain in the hospital. Hospital executives and healthcare professionals can benefit from accurate LOS prediction in a number of ways, including resource allocation optimization, discharge planning, and patient outcomes improvement. A patient's LOS can be impacted by a number of variables, including age, comorbidities, the severity of their disease or injury, and social and environmental circumstances. These variables can be used to create predictive models that calculate a patient's anticipated LOS. Models for forecasting can be created using machine learning techniques including LR,KNN, DT, RF and NB.

Healthcare readmissions to hospitals are a frequent and expensive problem. The term "readmission" describes a patient who, after being released from their prior hospitalization, returns to the facility within a specific time frame. Unexpected readmissions can lead to higher healthcare costs, lower satisfaction among patients, and worse patient outcomes. In order to reduce readmissions, healthcare providers can utilize predictive modeling to identify individuals who have a high risk of doing so. Predictive models for readmission can be created using machine learning algorithms including models for forecasting can be created using machine learning techniques including LR, DT, RF, and NB.

One of the top causes of death globally and an important threat to public health is heart disease. It covers a wide spectrum of heart ailments include arrhythmias, heart failure, and coronary artery disease. Numerous factors, such as genetics, way of life, and underlying medical conditions, can cause these conditions. Machine learning algorithms including models for forecasting can be created using machine learning techniques including LR, DT, RF, and NB

A. Length of Stay Data Analysis

```
Data columns (total 18 columns):
 #   Column                              Non-Null Count   Dtype
---  ------                              --------------   -----
 0   case_id                             318438 non-null  int64
 1   Hospital_code                       318438 non-null  int64
 2   Hospital_type_code                  318438 non-null  object
 3   City_Code_Hospital                  318438 non-null  int64
 4   Hospital_region_code                318438 non-null  object
 5   Available Extra Rooms in Hospital   318438 non-null  int64
 6   Department                          318438 non-null  object
 7   Ward_Type                           318438 non-null  object
 8   Ward_Facility_Code                  318438 non-null  object
 9   Bed Grade                           318325 non-null  float64
 10  patientid                           318438 non-null  int64
 11  City_Code_Patient                   313906 non-null  float64
 12  Type of Admission                   318438 non-null  object
 13  Severity of Illness                 318438 non-null  object
 14  Visitors with Patient               318438 non-null  int64
 15  Age                                 318438 non-null  object
 16  Admission_Deposit                   318438 non-null  float64
 17  Stay                                318438 non-null  object
dtypes: float64(3), int64(6), object(9)
memory usage: 43.7+ MB
```

Fig. 1. Data Analysis

The following columns are present in the dataset:
caseid:Case ID for a hospital patient (Nominal). Hospital-

specific identification code (Nominal) Hospitaltypecode : Special code identifying the hospital type (Nominal) CityCodeHospital: The Hospital's City Code (Nominal) Hospital Region Code: (Nominal) Hospital Region Code Number of additional hospital rooms that are now available (in numbers) Department: (Nominal) Department ignoring the case WardType: Nominal code for the Ward type The Ward Facility's code is WardFacilityCode (Nominal). Bed Grade: The orderly condition of the beds in the hospital. Unique Patient Id (Nominal): patient-id CityCodePatient: The patient's city code (Nominal). Admission Type: Hospital-registered (Nominal) Admission Type Condition Severity: Condition severity as reported at the time of admission (Ordinal). Patients' Visitors: The patient's (numerical) visitors' number. Age: The patient's age (in years). Deposit made at the time of admission; admissiondeposit. Days the patient stays in the hospital (target variable)

B. Readmitted Data Analysis

```
Data columns (total 23 columns):
 #   Column                    Non-Null Count   Dtype
---  ------                    --------------   -----
 0   encounter_id              101766 non-null  int64
 1   patient_nbr               101766 non-null  int64
 2   gender                    101766 non-null  object
 3   age                       101766 non-null  int64
 4   weight                    101766 non-null  object
 5   admission_type_id         101766 non-null  int64
 6   discharge_disposition_id  101766 non-null  int64
 7   admission_source_id       101766 non-null  int64
 8   time_in_hospital          101766 non-null  int64
 9   num_lab_procedures        101766 non-null  int64
 10  num_procedures            101766 non-null  int64
 11  num_medications           101766 non-null  int64
 12  number_outpatient         101766 non-null  int64
 13  number_emergency          101766 non-null  int64
 14  number_inpatient          101766 non-null  int64
 15  number_diagnoses          101766 non-null  int64
 16  insulin                   101766 non-null  object
 17  glipizide-metformin       101766 non-null  object
 18  glimepiride-pioglitazone  101766 non-null  object
 19  metformin-rosiglitazone   101766 non-null  object
 20  metformin-pioglitazone    101766 non-null  object
 21  diabetesMed               101766 non-null  object
 22  readmitted                101766 non-null  object
```

Fig. 2. Data Analysis

The following columns are present in the dataset:
hospitaladmission is an example of an encounter with an encounterid. a patient's individual identification number, or patientnbr gender: the patient's gender age: the patient's age range patient's weight, which may not be available for all patients. An identification of the type of admission (such as emergency or elective), admissiontypeid: admission-id. dischargedispositionid: a code that designates how the patient will be discharged (e.g., sent home or sent to another hospital).An identification number for the source of the patient's admittance (such as a doctor's recommendation or an emergency room). numberlabprocedures: the number of lab tests the patient underwent during their stay numberprocedures: the number of non-lab procedures the patient underwent during their stay timeinhospital: the number of days the patient remained in the hospital. nummedications: the quantity of prescription drugs given to the patient during their stay.the number of outpatient visits the patient made within the calendar year prior to their admissionThe patient's total number of emergency visits during the year prior to admissionthe patient's total number of inpatient visits during the calendar year prior to their admissiondiagnoses: the total number of diagnoses the patient received while hospitalized whether the patient received an insulin prescription during their stay whether the patient received a prescription for glipizide and metformin together during their stay whether the patient received a prescription for glimepiride and pioglitazone together while they were there. whether the patient received a prescription for metformin and rosiglitazone together while they were there whether the patient received a prescription for metformin and pioglitazone together while they were there if the patient received a prescription for diabetes medication while they were there (diabetesMed) Whether the patient was readmitted to the hospital again in less than 30 days

C. Heart Disease Data Analysis

```
Data columns (total 18 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   HeartDisease      319795 non-null  object
 1   BMI               319795 non-null  float64
 2   Smoking           319795 non-null  object
 3   AlcoholDrinking   319795 non-null  object
 4   Stroke            319795 non-null  object
 5   PhysicalHealth    319795 non-null  float64
 6   MentalHealth      319795 non-null  float64
 7   DiffWalking       319795 non-null  object
 8   Sex               319795 non-null  object
 9   AgeCategory       319795 non-null  object
 10  Race              319795 non-null  object
 11  Diabetic          319795 non-null  object
 12  PhysicalActivity  319795 non-null  object
 13  GenHealth         319795 non-null  object
 14  SleepTime         319795 non-null  float64
 15  Asthma            319795 non-null  object
 16  KidneyDisease     319795 non-null  object
 17  SkinCancer        319795 non-null  object
dtypes: float64(4), object(14)
```

Fig. 3. Data Analysis

18 columns (features) in a data-set without any indication of the number of rows or the data type for each column. HeartDisease: Whether a patient has heart disease or not. We will be using the other variables to forecast this target variable. BMI: The patient's body mass index. Smoking: Whether or not the patient smokes. AlcoholDrinking: The patient's alcohol consumption. Stroke: Whether or not the patient has experienced a stroke. A measurement of the patient's physical health. A measurement of the patient's mental health is called mentalHealth. Diff-Walking: A measurement of the patient's walking difficulties. Sex: The patient's sex. AgeCategory: The patient's age group. Race: The patient's race. Diabetic: The presence or absence of diabetes in the patient. Physical activity is a metric for how physically active a patient is. GenHealth: measure of the patient's general health. SleepTime: The typical daily sleep duration for the patient. Whether or not the patient has asthma. The presence or absence of renal disease in the patient. Whether or not the patient has ever had skin cancer.

II. RESEARCH QUESTIONS

A. Predicting the Length of Stay in hospitals?

B. Predicting age-wise readmitted patients in hospitals?

C. Predicting Heart Disease patients in hospitals?

The research questions mentioned above will reveal the LOS, readmitted patients, Heart Disease patients in hospitals.

III. Motivation

For patients and healthcare systems alike, the ability to forecast hospital length of stay, cardiac disease, and readmission have important practical implications. Patients can better plan their post-hospitalization treatment, such as discharge planning and home health care, if the length of their stay can be accurately predicted. Early detection and identification of cardiac disease can result in prompt treatment and improved results. Accurate readmission prediction can aid hospitals and healthcare systems in identifying areas where treatment quality can be enhanced, resulting in cost savings and improved patient outcomes. Overall, precise forecasting of these variables can raise the standard of care and lighten the load on the healthcare system.

Healthcare professionals can more effectively plan and deploy resources, enhance patient outcomes, and cut costs by predicting how long patients will stay in the hospital. Machine learning models can be trained to accurately forecast the duration of stay for a specific patient by studying patient variables including age, sickness severity, type of admission, and length of stay. By using this information, healthcare professionals can spot patients who are most likely to require an extended stay and take the necessary steps to manage their treatment successfully.

Readmissions to the hospital place a heavy load on the healthcare system and can have a negative impact on patient outcomes. It is possible to identify patients who are at readmission using predictive modeling approaches, enabling healthcare professionals to take early action and enhance patient outcomes. Machine learning models can be taught to predict the chance of readmission for a specific patient by studying patient data such as age, length of stay, type of admission, and comorbidities. By identifying high-risk patients and implementing focused interventions, healthcare practitioners can lower the chance of readmission.

The biggest cause of illness and mortality in the globe is heart disease. To identify patients who are at risk of developing heart disease and to forecast the likelihood of unfavorable outcomes including heart attacks, strokes, and hospital readmissions, predictive modeling techniques can be applied to digital medical record data. To identify patients who are at high risk for developing heart disease or suffering negative outcomes, machine learning models can be trained on a variety of clinical and demographic variables, which includes gender, age, habit of smoking, cholesterol level, blood pressure, and family history of heart disease.

IV. Related Work

A review of the literature on the predictors of length of stay in the intensive care unit is presented in the paper of Critical Care. The length of stay in the ED was analyzed in multiple publications, and the authors found significant factors, including age, illness severity, and entry diagnosis, among others. The Journal of Critical Care article "Heart disease and LOS in the intensive care unit" is another pertinent one. The goal of this study was to determine what factors affect how long heart disease patients stay in the intensive care unit. studies that use DT, LR as machine learning models to predict the length of stay in hospitals. The authors point out that although earlier research has shown encouraging findings, more work is still needed, particularly when it comes to the application of neural networks.[1]

The goal of the 2009 study, which was published in JAMA Internal Medicine, was to investigate the relationship between hospital readmission rates and clinical outcomes, such as mortality and resource usage, among American Medicare beneficiaries. Between July 2003 and June 2004, over 6 million Medicare beneficiaries were discharged from acute care hospitals. The authors studied the data from these beneficiaries and discovered that those who were readmitted within 30 days had higher mortality rates and consumed more healthcare resources. They came to the conclusion that decreasing hospital readmissions could enhance patient outcomes and lower medical expenses.The study emphasizes the significance of anticipating and preventing hospital readmissions because they might have detrimental effects on patient health and financial resources for healthcare. Healthcare practitioners can better target interventions to lower the incidence of readmission and enhance patient outcomes by identifying patients who are at high risk for readmission.[2]

The article offers thorough details on risk factors, outcomes, and healthcare utilization, as well as the prevalence and incidence of heart disease and stroke. Additionally, it contains details on how different genders and ethnic people with different statuses , geographical locations, and people with different access to care differ in incidence, mortality, and the outcomes of heart disease and stroke. The article's discussion of measures to enhance healthcare delivery and lessen disparities, as well as the significance of preventing and controlling heart disease and stroke, comes to a conclusion. In conclusion, this article offers a useful resource for academics, medical professionals, and decision-makers who are tackling the burden of heart disease and stroke in the US. [3]

In a study, the connections between hospital readmissions, stroke, and myocardial infarction (MI) are examined. According to the study, readmission rates were high among stroke and MI patients, with a sizable portion of patients returning to the hospital within 30 days of their first hospitalization. Age, comorbidities, and hospital features were all identified in the study as predictors of readmission.The authors stress the need of identifying patients who are at high risk for readmission and putting preventative measures in place, like better discharge planning and care coordination. The study emphasizes the need for additional investigation to create practical plans for lowering hospital readmissions in this patient population.[4]

The study paper you've posted, written by Lee et al, explains how to leverage data from electronic health records (EHRs) to forecast longer lengths of hospital stays (LOS) for admitted patients. The authors created a prediction model using multiple demographic, clinical, and laboratory characteristics after conducting a retrospective research using EHR data from a

significant academic medical facility. According to the study, the prediction model that used EHR data could correctly predict admitted patients' longer LOS, which would enable earlier intervention and better patient outcomes. According to the authors, this methodology might be utilized to create real-time clinical decision support systems that would assist healthcare professionals in managing hospital resources and enhancing patient care. [5]

In a tertiary hospital in Nigeria, the study looked for prevalence of depression and anxiety symptoms in patients with heart disease. 380 heart disease patients were included in the study using a cross-sectional design by the researchers. They used the Hospital Anxiety and Depression Scale (HADS) to gather data, and descriptive and inferential statistics were used to analyze the results. According to the findings, 37.6 % of the patients experienced symptoms of anxiety, and 43.9% of the patients had signs of sadness. The study also discovered that depressive symptoms were more common in patients with cardiac disease who had had it for longer than five years. The study's findings that depression and anxiety symptoms were common among cardiac patients in Nigeria led the researchers to urge routine testing for these conditions in these patients in order to improve their quality of life. [6]

This paper conducts a thorough evaluation of the literature on predicting readmissions to hospitals. It comprises research that employed statistical models to forecast readmissions and were published between January 2000 and June 2013. The evaluation looks at how the research were conducted, the predictors they employed, and how well the models worked. In order to create and validate precise and dependable prediction models for hospital readmissions, further research is required, as it is concluded that there is a broad range in the methods and predictors utilized in the studies. [7]

The LACE index, the Readmission Risk Score (RRS), and a hybrid model were the three models that the study compared for predicting hospital readmission among patients with heart failure. Four factors make up the LACE index: length of stay, entrance acuity, comorbidities, and ER visits in the six months before to admission. 13 variables that are related to patient demographics, clinical variables, and hospitalization features are included in the RRS. The LACE index and RRS are combined in the hybrid model. According to the study, among patients with heart failure, the hybrid model showed the highest prediction accuracy for hospital readmission. According to the study, risk adjustment models can be helpful for identifying patients who are more likely to require hospital readmission and for focusing readmission prevention efforts. [8]

The authors give an overview of the various factors that contribute to hospital readmissions, including patient factors like age, comorbidities, and self-care behaviors, as well as system-level factors like hospital discharge procedures and access to outpatient care, in the article that highlights how frequent hospital readmissions are among patients with diabetes. The paper also covers possible strategies to lessen hospital readmissions among diabetic patients, such as enhancing coordination of care and communication between inpatient and outpatient healthcare professionals and putting transitional care programs in place. The research underscores the need for comprehensive, multifaceted strategies to prevent hospital readmissions and enhance patient outcomes, and overall it offers insightful information on the complicated problem of hospital readmissions among diabetic patients. [9]

Longer lengths of stay were linked to a decreased rate of readmission for heart failure, according to the study, which included data from 129 hospitals in 18 states. The authors hypothesize that this might be because the lengthier stay required more intensive monitoring and therapy. They also point out that prolonged hospital stays raise healthcare expenses and may not always be beneficial or essential for patients. The study's overall findings emphasize the importance of creating personalized treatment programs and giving careful thought to the length of stay for heart failure patients.[10]

The paper explores the connections among substance users in an inner-city population, social functioning, emotional repression, and avoidance, as well as indicators of posttraumatic stress disorder (PTSD). According to the study, emotional avoidance and repression may be factors in the link between worse social functioning and higher levels of PTSD symptoms.[11]

The use of machine learning algorithms to assess the risk of heart disease is reviewed in the article. The authors stress the value of early cardiac disease identification in preventing complications and enhancing patient outcomes. They give an overview of different machine learning methods and talk about how they might be used to predict cardiac disease. Examples of these methods include DT, LG. The difficulties in creating and putting these models into clinical practice are also covered in the paper. The article offers examination of the state of machine learning techniques for heart disease prediction right now.[12]

V. METHODOLOGY

A. Data-sets Description

Kaggle is the platform used for data science competitions. Users can also access a variety of databases for study and teaching. Unfortunately, you did not specify which Kaggle dataset you were referring to. This dataset contains information on patients who were admitted to a hospital, including the patient's age, gender, race, diagnosis, medical history, length of stay, and whether they were readmitted within 30 days of discharge. Over 130 US hospitals provided the information between 2015 and 2016. The dataset aims to predict a patient's duration of stay and likelihood of readmission using their medical history and demographic data. 40 columns and more than 100,000 rows make up the dataset, which includes both numerical and categorical variables

B. Technology Used

Logistic regression

For binary classification issues, the common approach for machine learning is logistic regression. Logistic regression can be used in an emergency room to forecast a patient's

chances of being admitted or discharged based on their medical history, vital signs, and other pertinent variables.

K Nearest Neighbour

A well-liked approach in machine learning for classification and regression issues is K-nearest neighbor (KNN). KNN can be used in an emergency room to forecast a patient's chances of being admitted or discharged based on their medical history, vital signs, and other pertinent data.

Decision Tree

Decision trees are a popular machine learning method for problems involving classification and regression. A decision tree can be used in the emergency department to predict whether a patient will be admitted or discharged based on their medical history, vital signs, and other relevant information. Decision trees are a popular machine learning method for problems involving classification and regression. A decision tree can be used in the emergency department to predict whether a patient will be admitted or discharged based on their medical history, vital signs, and other relevant information.

Random forest (RF) algorithm

The famous ensemble learning method known as random forest (RF) is used in machine learning to solve classification and regression problems. It is a good idea to have a backup strategy in case the backup fails.

Naïve Bayes

A classification technique called Naive Bayes operates by probabilistically examining the connections between input variables and output classifications. Based on a patient's medical vital signs, history, and other relevant data, Nave Bayes can be used to forecast whether a patient would be hospitalized or discharged in the context of an emergency department.

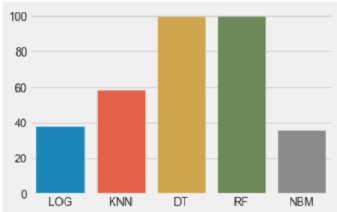

Fig. 4. Model 1

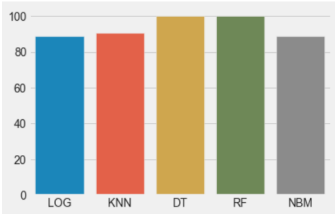

Fig. 5. Model 2

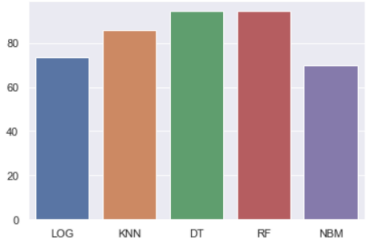

Fig. 6. Model 3

C. Project Flow

Exploratory Data Analysis

The process of looking at and visualizing the data to comprehend its key features, find trends, connections, and insights, as well as to spot any potential problems with the data. By computing measures of central tendency and variability, analyzing the distribution of the data, and producing graphs and charts to visualize the data, EDA entails utilizing statistical and visualization techniques to condense the data.[Fig.1][Fig.2][Fig.3]

Distribution of target feature

We must first determine which feature is the target feature in order to visualize the distribution of the target feature. It is unclear which characteristic in the provided dataset is the desired feature. Please elaborate on the dataset and identify the goal feature in the dataset.

Check Null Values,Duplicated values,Data Analysis Visualization.

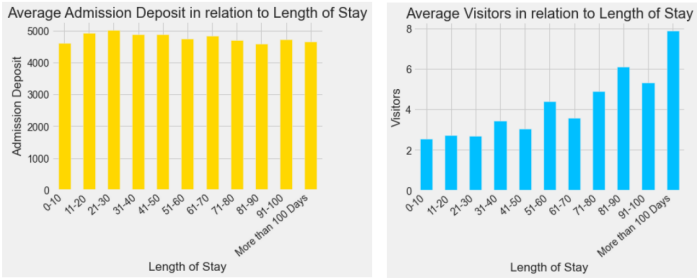

Fig. 7. Visitors & Admission Deposit LOS

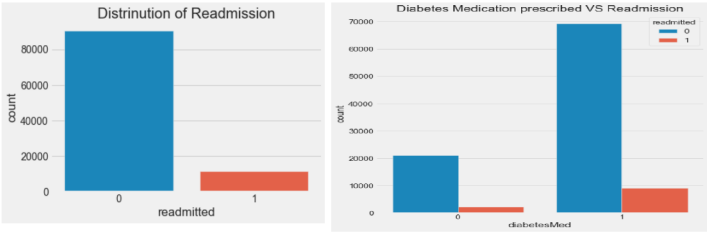

Fig. 8. Distrinution of Readmission & Diabetes Medication prescribed VS Readmission

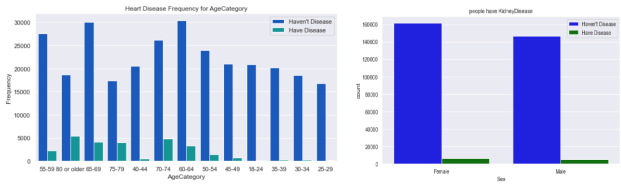

Fig. 9. Have Disease by Sex & Age

VI. RESULT AND EVALUATION

Finding the correct insight is crucial for data analysts since it not only saves time but also reduces the possibility of errors arising from improper data analysis. Sometimes, even information that is unrelated to our research topics or business issues might be used to solve problems that may arise in the future or even right now. So let's look at some key conclusions we came to from our study before we address our research questions.

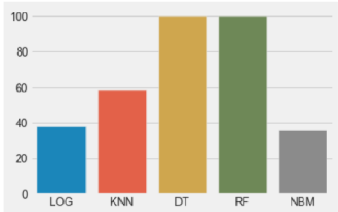

Fig. 10. Predicting the Length of Stay

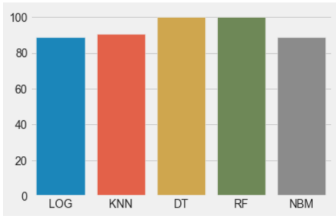

Fig. 11. Predicting the readmitted patients in hospitals

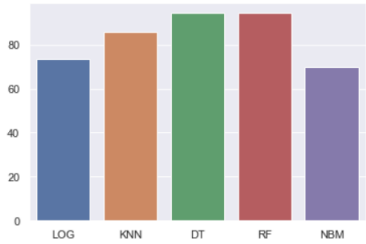

Fig. 12. Predicting the Heart Disease in hospitals

VII. RESEARCH QUESTION

A. *Predicting the Length of Stay in hospitals?*

Model choice using the Best Subsets The optimal combination of predictor variables that can accurately predict the result variable is found using an algorithm, which is a statistical technique. The algorithm identifies the subset of variables that best fits the model by examining all feasible subsets of predictor variables. In Fig 13 describe the overall admission in

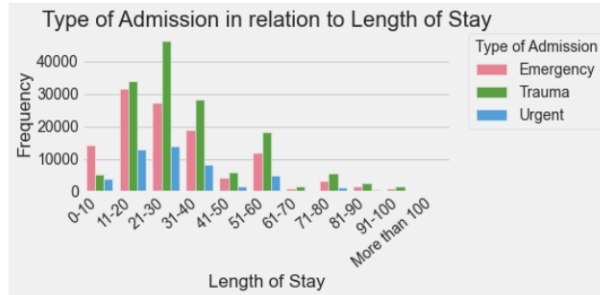

Fig. 13. Type of Admission in relation to Length of Stay

hospital and type of admission. Fig 10 explain the prediction for LOS using LR, KNN , DT, RF, NB.The best model is DT,RF and the worst model is NB.

B. *Predicting age-wise readmitted patients in hospitals?*

Utilizing the Best Subsets, choose a model An algorithm, which is a statistical approach, is used to identify the best possible set of predictor variables that may successfully predict the outcome variable. By looking at all viable subsets of predictor variables, the method determines the subset of variables that best fits the model. In Fig 14 describe the overall

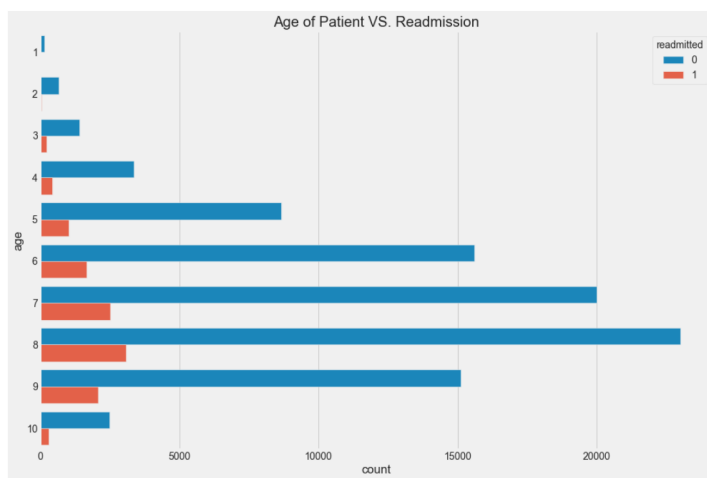

Fig. 14. Age of Patient VS. Readmission

age group readmitted patients in hospital. Fig 11 explain the prediction for LOS using LR, KNN , DT, RF, NB.The best model is DT,RF and the worst model is NB.

C. *Predicting Heart Disease patients in hospitals?*

Select a model using the Best Subsets. To find the best feasible set of predictor variables that may correctly predict the result variable, an algorithm—a statistical approach—is applied. The method identifies the subset of variables that best fits the model by considering all feasible subsets of predictor

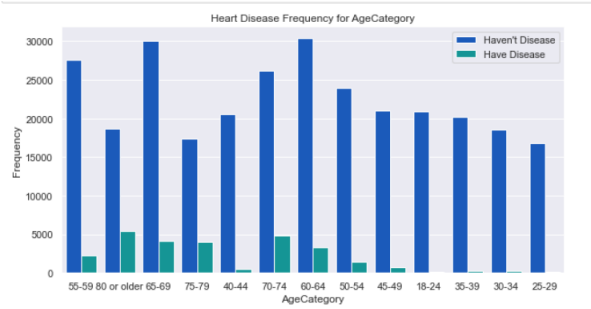

Fig. 15. Heart Disease Frequency for AgeCategory

variables. In Fig 15 describe the overall age group Heart Disease patients in hospital. Fig 12 explain the prediction for LOS using LR, KNN , DT, RF, NB.The best model is DT,RF and the worst model is NB.

VIII. Conclusion and Future Scope

In this research we have evaluated multiple machine learning model and we found that the best model is DT then RF then LR then KNN then NB at this order and worst model is NB model.

Creating a predictive model to recognize patients who run the risk of being in the hospital for an extended period of time or being readmitted could result in better resource allocation and better patient outcomes. comparing the efficiency of various medical interventions and therapies in shortening stay times and readmissions. evaluating potential strategies to address these inequities, as well as the influence of social determinants of health (such as poverty, education, and housing) on length of stay and readmissions. examining the use of natural language processing and machine learning algorithms to extract helpful information from electronic health records (EHRs) and enhance patient outcomes. investigating how hospital readmissions and stay length are affected by healthcare laws and regulations and offering suggestions for reforms that could benefit patients. tracking the health outcomes of patients after they leave the hospital and identifying the elements that lead to a good recovery and a decreased need for readmission. examining how well telemedicine and other forms of remote monitoring can be used to decrease readmissions to hospitals and enhance patient outcomes.

IX. References

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